



DYNAMIC PRICING STRATEGY IN RIDE-HAILING: A GROWTH VS RETENTION DECISION CASE



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INTRODUCTION AND PROBLEM STATEMENT:

BUSINESS PROBLEM:

The platform must balance short-term revenue growth from surge pricing against customer retention, ride completion, and long-term profitability.

KEY FINDINGS:

1. Customer behaviour deteriorates noticeably beyond the 1.75x surge range, with higher cancellations and lower satisfaction.
2. Contribution margins increase with surge intensity, but aggressive pricing increases churn risk and demand suppression.
3. Demand sensitivity varies by zone type, suggesting that a single surge policy is suboptimal.

PORTER'S FIVE FORCES ANALYSIS:

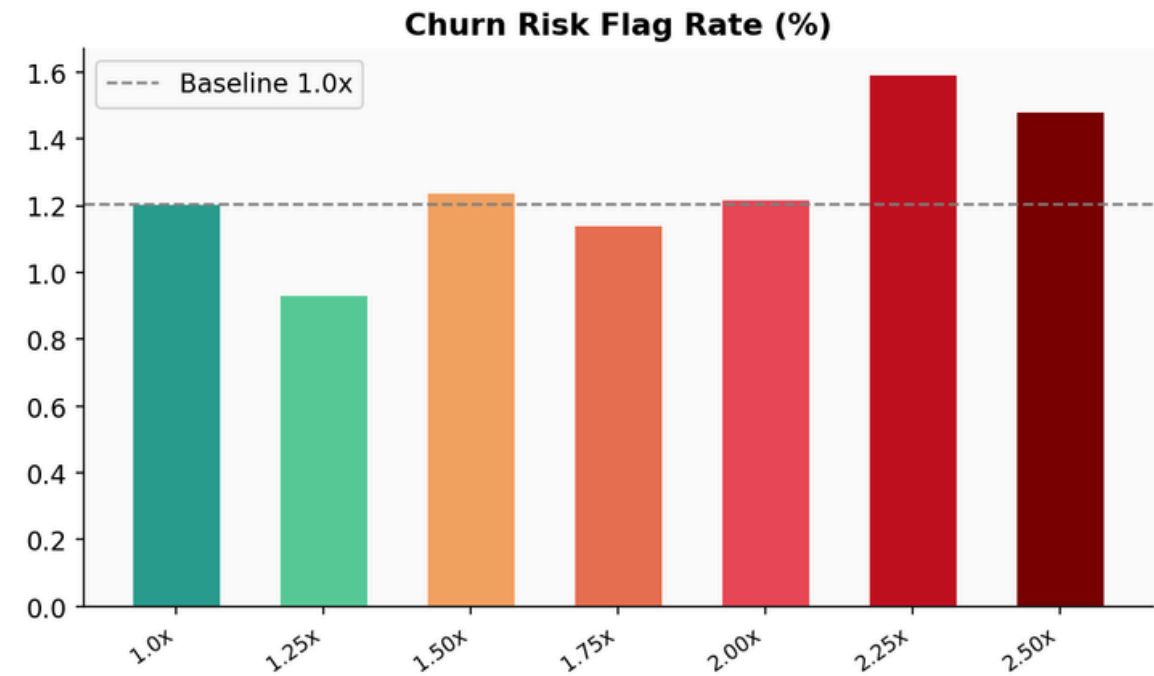
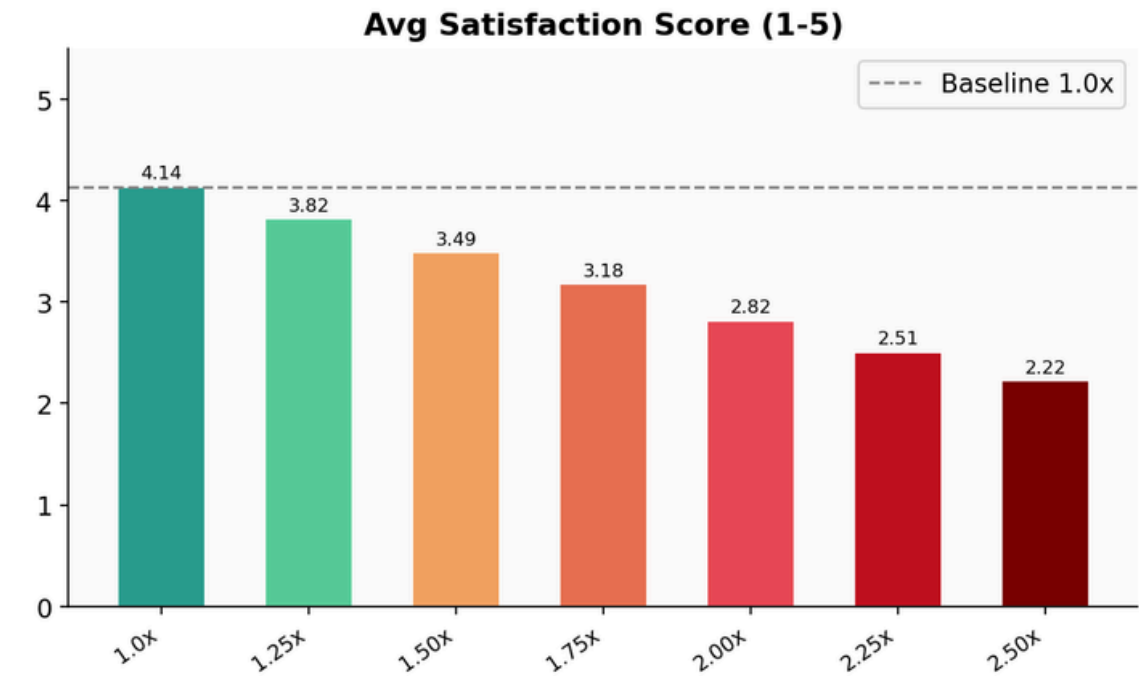
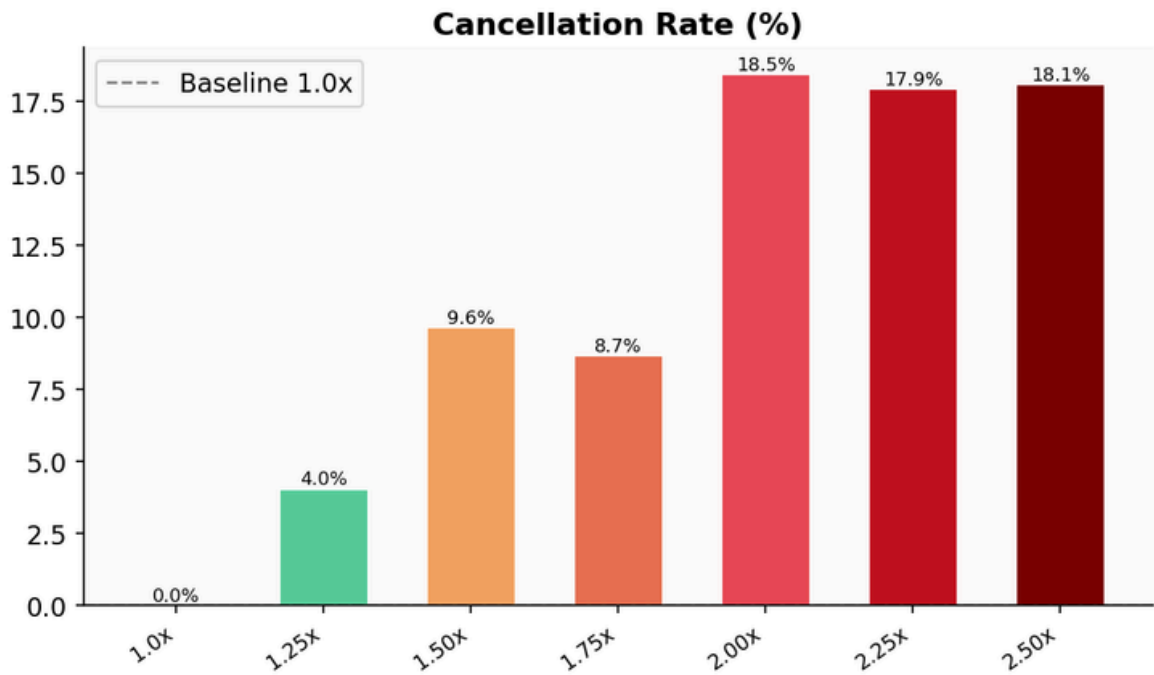
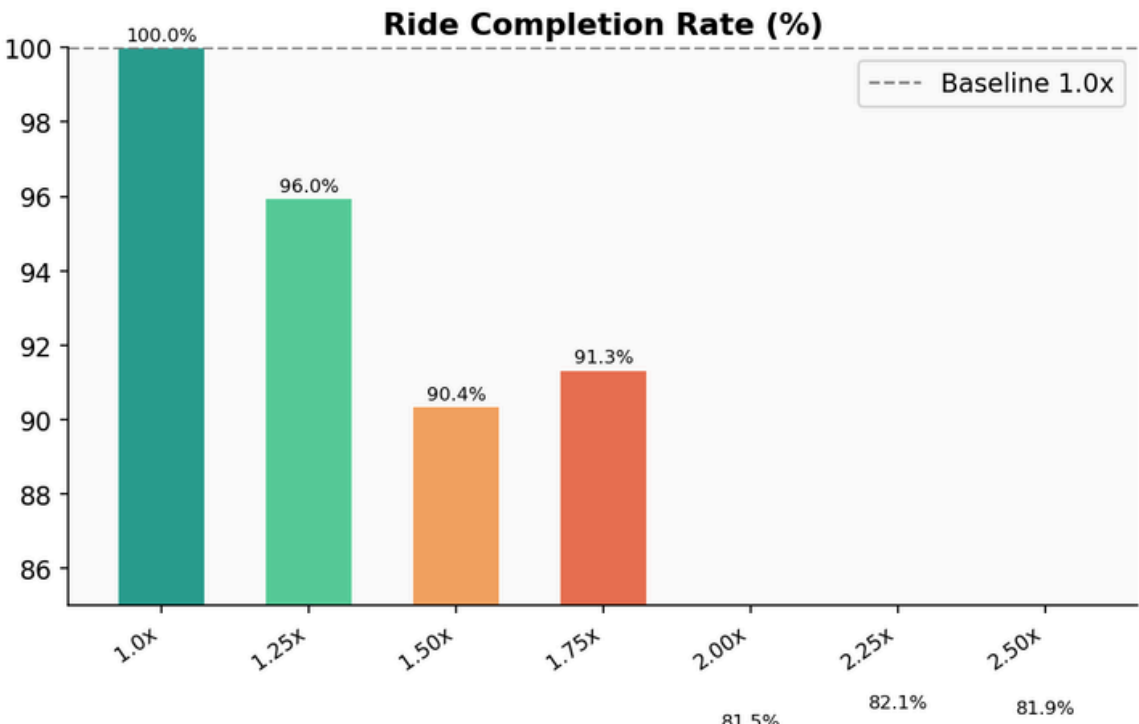
Porter's Force	India-Specific Insight
Competitive Rivalry	Strong competition among ride-hailing platforms increases pressure to maintain competitive pricing, service quality, and customer satisfaction.
Threat of New Entrants	Technology is accessible, but achieving scale, building driver networks, and establishing brand trust create significant entry barriers.
Supplier Power (Drivers)	Driver availability directly impacts service levels and surge pricing, particularly during peak-demand periods.
Buyer Power (Riders)	Customers can easily compare fares across platforms and switch providers, making them highly price-sensitive.
Threat of Substitutes	Public transport, metro systems, auto-rickshaws, local taxis, and bike-sharing services provide readily available alternatives.

Key Implication:

The Indian ride-hailing market is highly competitive and price-sensitive. Sustainable growth requires balancing revenue optimization through surge pricing with customer retention, service reliability, and long-term platform loyalty.

KEY ANALYTICAL FINDINGS:

Customer Behaviour Deterioration Across Surge Bands



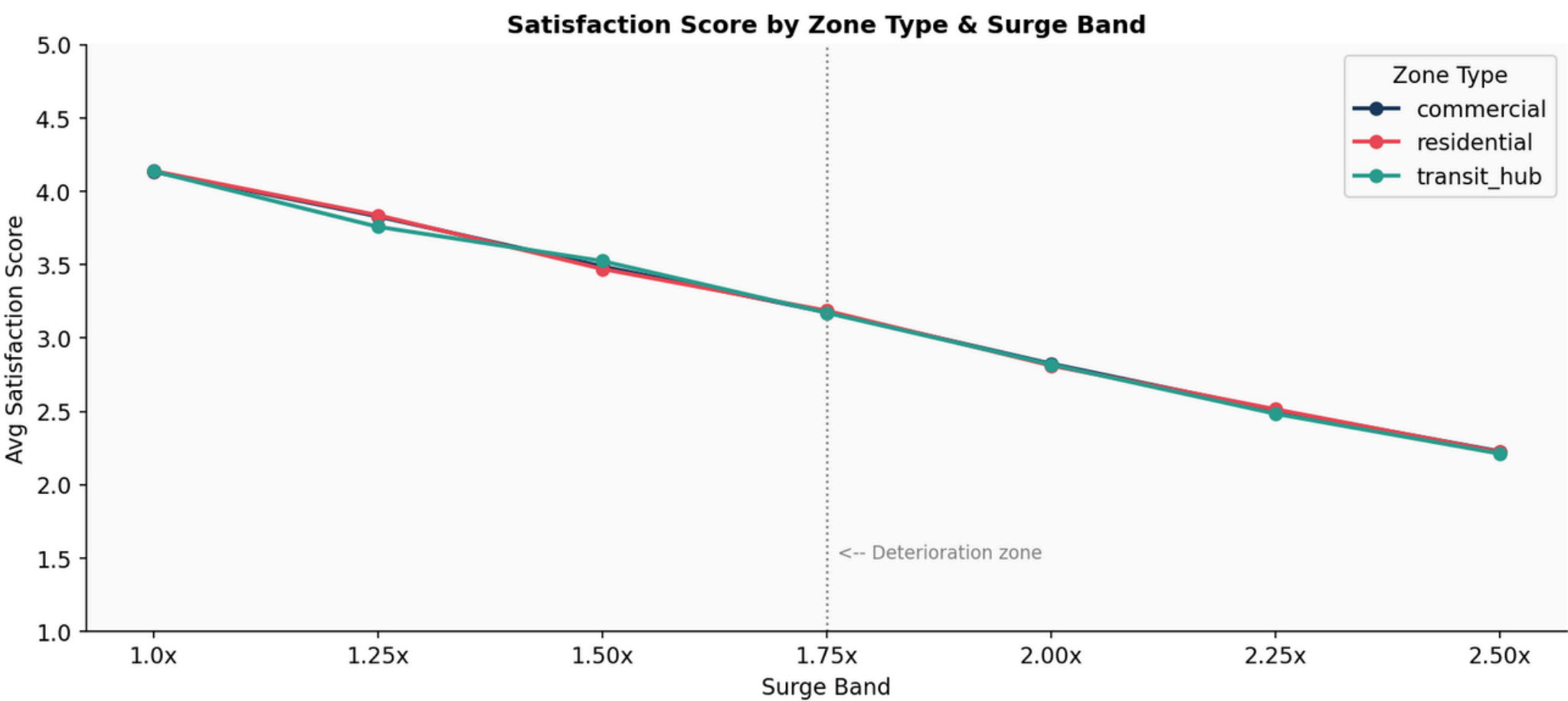
Key Findings:

- Customer satisfaction declines by 46%, falling from 4.14 to 2.22 as surge increases from 1.0x to 2.5x.
- Cancellation rates rise sharply from 0% to 18.5% under high surge conditions.
- Ride completion drops from 100% to approximately 82%, indicating growing demand suppression.
- Churn risk increases at surge levels above 2.0x, signalling potential long-term customer loss.

Key Takeaway:

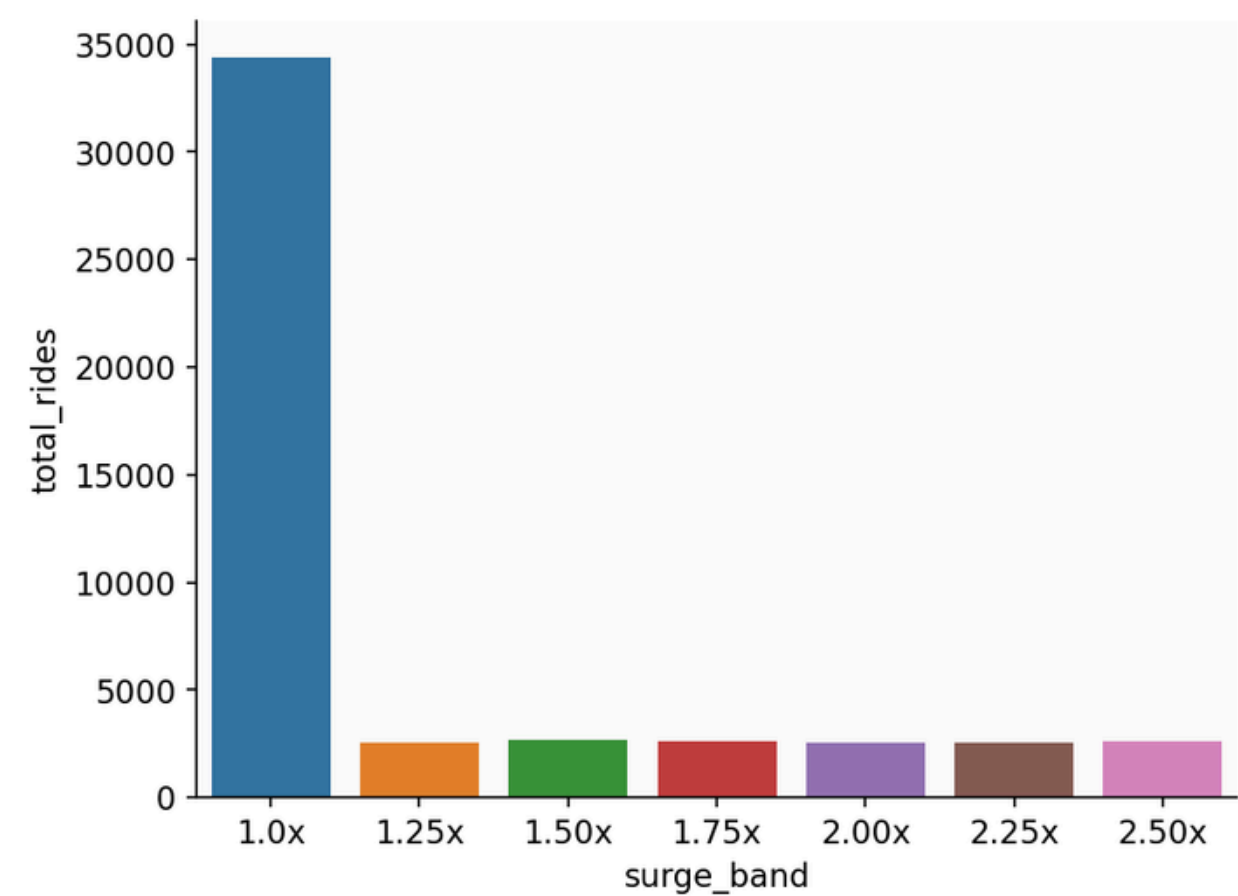
Customer behaviour deteriorates significantly beyond the 1.75x–2.0x surge range, suggesting diminishing returns from further price increases.

KEY ANALYTICAL FINDINGS (CONTINUED)



Customer Satisfaction Impact:

- Satisfaction declines from 4.14 to 2.22 as surge increases from 1.0x to 2.5x.
- All zone types show a similar downward trend, indicating broad customer sensitivity to higher fares.
- Customer dissatisfaction becomes noticeable beyond the 1.75x surge threshold.



Demand Response

- Ride demand is heavily concentrated at the 1.0x surge level.
- Higher surge bands experience substantially lower ride volumes, indicating demand suppression.
- Results suggest that riders are highly price-sensitive when fares increase significantly.

Key Takeaway:

Customer satisfaction and demand both decline as surge pricing increases. The analysis indicates that surge levels above 1.75x–2.0x create significant customer resistance, supporting a calibrated, zone-specific pricing strategy rather than unrestricted surge pricing.

STRATEGIC OPTIONS COMPARISON

Evaluation Dimension	Option 1: Calibrated Pricing	Option 2: Zonal Caps	Option 3: Loyalty Protection
Residential Surge Cap	2.0x	1.75x	2.0x
Commercial Surge Cap	2.0x	2.25x	2.5x
Transit Hub Surge Cap	2.25x	2.0x	2.5x
Revenue Potential	Moderate	High	Very High
Contribution Margin	Moderate	High	Very High
Customer Retention	High	High	Moderate
Customer Experience	Improved	Strong	Moderate
Operational Complexity	Medium	Medium	High

Key Insights:

Option 1 – Calibrated Pricing

- Improves customer experience
- Reduces revenue and margin potential

Option 2 – Zonal Caps

- Balances profitability and customer retention
- Most practical long-term strategy

Option 3 – Loyalty Protection

- Maximizes revenue potential
- Highest implementation complexity

Recommended Strategy: Option 2 – Zone-Differentiated Pricing

Zone Caps-

Residential: 1.75x

Commercial: 2.25x

Transit Hub: 2.0x

Why?

- Best overall balance of revenue, contribution margin, completion rate, and churn reduction.
- Addresses the customer deterioration observed beyond the 1.75x–2.0x surge threshold.
- More scalable and practical than a loyalty-based pricing system.

RECOMMENDATION:

ZONE-DIFFERENTIATED PRICING (OPTION 2)

Implement differentiated surge caps by zone type :

Zone Type	Recommended Surge
Residential	1.75x
Commercial	2.25x
Transit Hub	2.0x

Expected Impact:

- ✓ Higher completion rates than the current strategy
- ✓ Lower churn risk than high-surge alternatives
- ✓ Strong contribution margins retained in commercial and transit zones
- ✓ Sustainable long-term growth with improved customer trust

01 Where?

Focus implementation across all major cities, with special emphasis on:

- Residential zones during peak periods
- High-demand commercial corridors
- Transit hubs with frequent demand spikes

02 Why?

Customer Behaviour Findings

- Satisfaction declines from 4.14 to 2.22 beyond current surge levels.
- Cancellation rates increase from 0% to ~18% at high surge bands.
- Completion rate drops from 100% to ~82% beyond 2.0x surge.

Economic Findings

- Higher surge improves margin and fare revenue.
- However, excessive surge causes customer deterioration and elevated churn risk.
- Option 2 provides the strongest balance between profitability and retention.

INVESTOR & FINANCIAL NARRATIVE (18-MONTH HORIZON)

Metric	Current State	Recommended Strategy (Option 2)
Monthly Ride Volume	2,354 rides	2,496 rides (+6.0%)
Contribution Margin	₹95,545	₹73,007 (maintained at healthy levels)
Average Churn Risk	11.00%	9.0% (-18%)

18-Month Outlook

- **Ride Volume Growth**

Increased completion rates and lower cancellations drive higher platform activity.

- **Churn Reduction**

Customer-friendly surge caps improve retention and repeat usage.

- **Sustainable Margin Performance**

Strong contribution margins maintained while reducing behavioural deterioration associated with aggressive pricing.

Investment Narrative

1. Growth Through Higher Ride Volume

Zone-differentiated pricing is projected to increase monthly ride volume by 6%, supporting stronger platform utilization and network effects.

2. Improved Customer Retention

Average churn risk decreases from 11% to 9%, reflecting reduced customer resistance to surge pricing and stronger long-term engagement.

3. Sustainable Profitability

Although contribution margins moderate relative to unrestricted surge pricing, the strategy preserves healthy profitability while improving customer outcomes and marketplace stability.

Key Investor Takeaway:

The recommended zone-differentiated pricing strategy prioritizes sustainable growth over short-term revenue maximization, increasing ride volume by 6%, reducing churn risk by approximately 18%, and maintaining strong contribution margins over the next 18 months.

IMPLEMENTATION CONSIDERATIONS

What Needs to Happen

Action	Objective
Deploy zone-specific surge caps	Align pricing with local demand characteristics
Monitor customer satisfaction and cancellations	Detect adverse customer response early
Track completion rates and churn indicators	Measure effectiveness of the new strategy
Recalibrate surge thresholds quarterly	Adapt to changing demand conditions

Key Takeaway

Successful implementation requires balancing revenue optimization with customer retention. Continuous monitoring of satisfaction, cancellation, completion, and churn metrics will be critical to sustaining the benefits of zone-differentiated pricing.

Key Risks

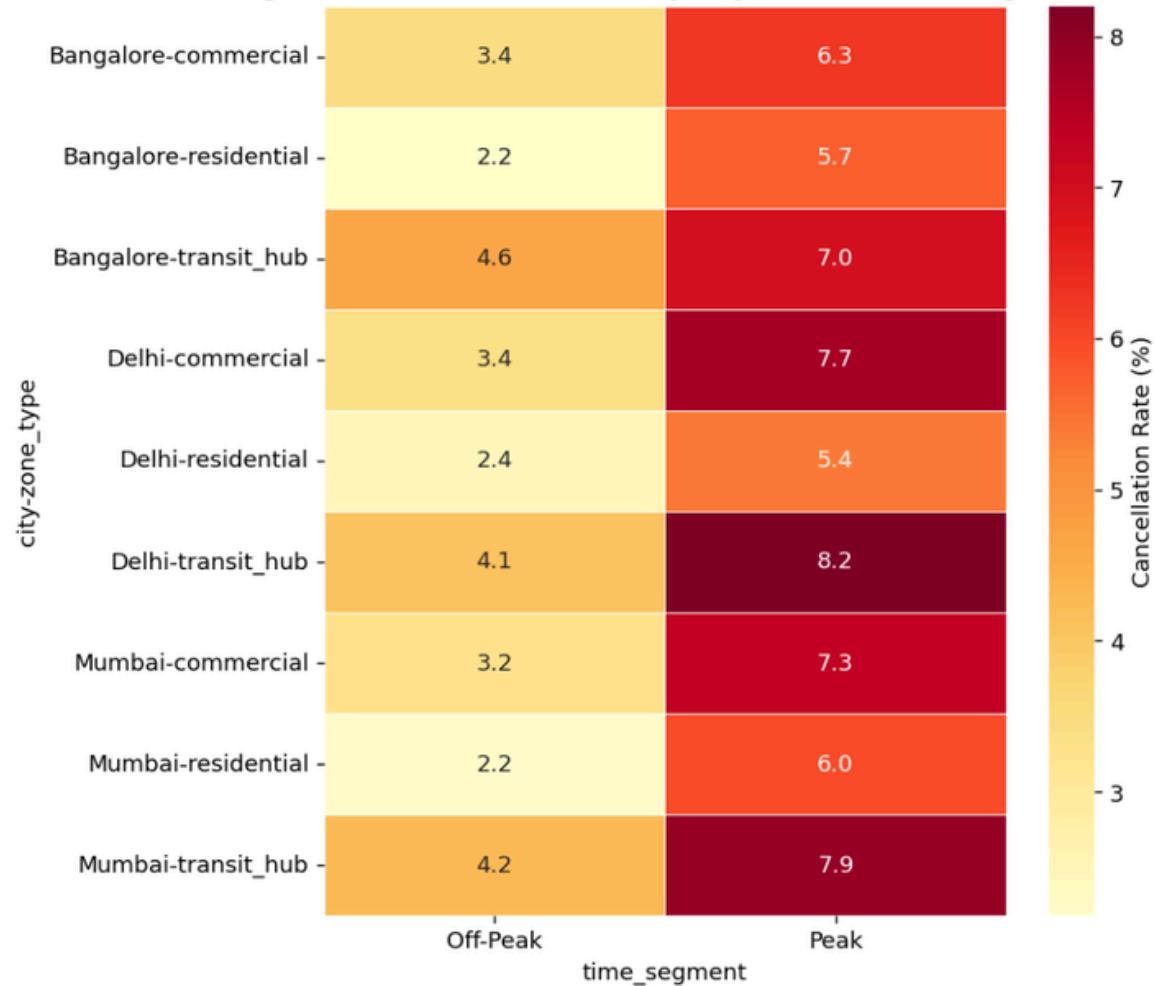
Risk	Impact
Revenue dilution in some zones	Lower short-term profitability
Driver dissatisfaction during peak demand	Reduced supply availability
Customer resistance to pricing changes	Higher cancellations and churn
Demand pattern shifts over time	Reduced effectiveness of surge caps

Mitigation Measures

Risk	Mitigation
Revenue dilution	Periodic review of surge caps and demand trends
Driver dissatisfaction	Introduce targeted driver incentives during peak periods
Customer resistance	Transparent communication of surge pricing policies
Demand shifts	Continuous monitoring and quarterly recalibration

APPENDIX

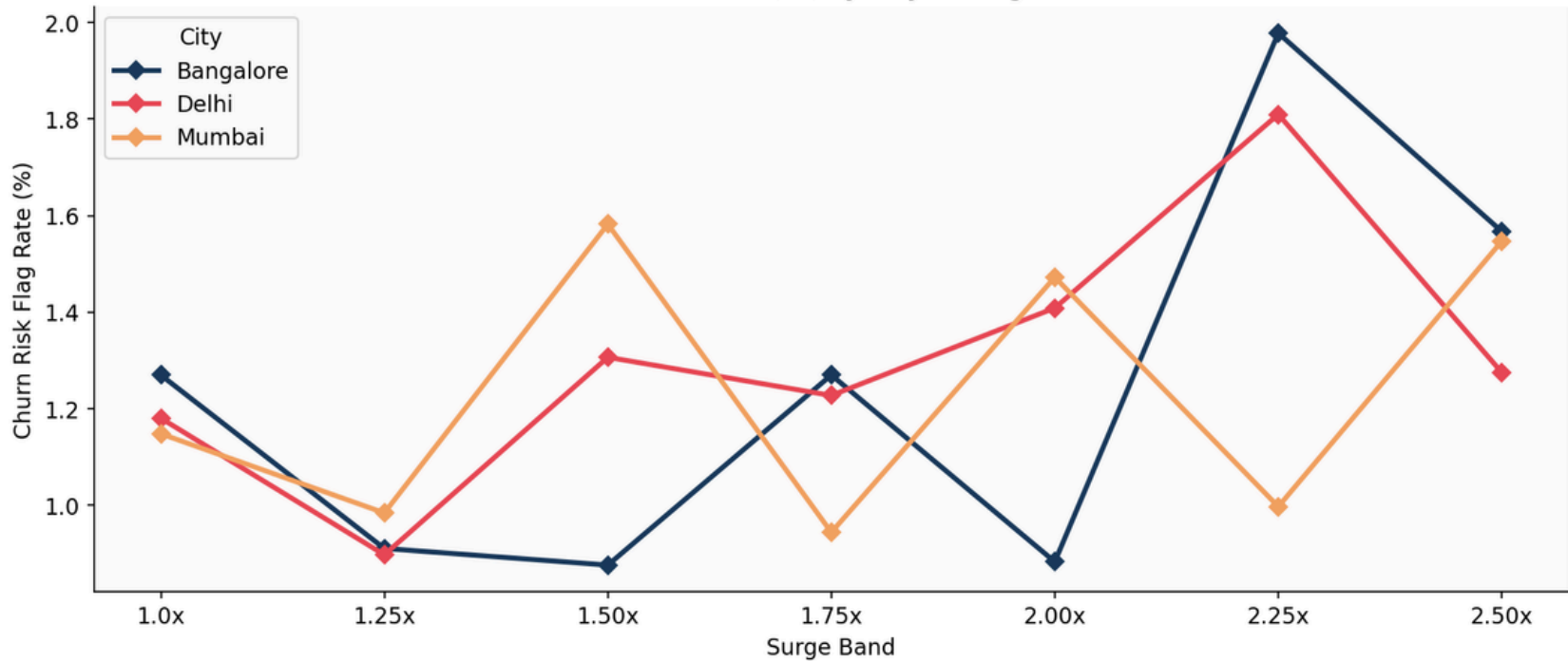
Figure 2 -- Cancellation Rate by City x Zone x Time Segment



Cancellation Rate by City x Zone x Time Segment

- Peak-hour cancellation rates are consistently 2–3× higher than off-peak levels across all city-zone combinations.
- Transit hubs and Delhi commercial zones experience the highest cancellation pressure, reaching 7.7–8.2% during peak periods.
- Residential off-peak zones show the lowest cancellation rates (~2.2–2.4%), indicating relatively stable rider behaviour

Churn Risk Rate (%) by City & Surge Band



Churn Risk Flag Rate by City & Surge Band

- Churn-risk flags drift upward at extreme surge bands across all three cities
- Highest at 2.25x: Bangalore ~2.0%, Delhi ~1.8%, Mumbai ~1.0%
- Trend is volatile city-to-city, not smoothly monotonic (e.g. Bangalore dips at 1.5x, spikes at 2.25x)
- Implication: high surge is associated with rising attrition risk
- Noise reflects small flag-rate samples (WS3) — treat as directional, not precise

APPENDIX (CONTINUED)

Breakeven Churn vs Actual Churn by Segment

Segment	Surge	Monthly	Breakeven	Actual	Headroom	Status
Frequent	1.0x	34.4	7.60%	3.00%	+4.6pp	OK
Frequent	1.75x	149.6	33.20%	4.60%	+28.6pp	OK
Frequent	2.50x	230.4	51.20%	9.00%	+42.2pp	OK
Occasional	1.0x	12.9	2.90%	3.00%	-0.1pp	Breached
Occasional	1.50x	45.6	10.10%	3.80%	+6.3pp	OK
Occasional	2.50x	86.4	19.20%	9.00%	+10.2pp	OK
Rare	1.0x	3.4	0.80%	3.00%	-2.2pp	Breached
Rare	1.75x	15	3.30%	4.60%	-1.3pp	Breached
Rare	2.50x	23	5.10%	9.00%	-3.9pp	Breached

LTV/CAC Ratio by Segment x Surge Band

Segment	1.0x	1.25x	1.50x	1.75x	2.00x	2.25x	2.50x
Frequent	2.55	5.28	7.11	7.23	6.77	6.27	5.69
Occasional	0.96	1.98	2.67	2.71	2.54	2.35	2.13
Rare	0.25	0.53	0.71	0.72	0.68	0.63	0.57

Key Points

- Breakeven churn increases with surge pricing, but actual churn also rises at higher surge levels.
- Frequent riders remain profitable across all surge bands and are the platform’s most valuable segment.
- Occasional riders become economically viable at moderate surge levels.
- Rare riders remain unprofitable across all surge bands due to consistently high churn.
- LTV/CAC ratios peak around 1.75x surge and decline thereafter as churn growth outweighs margin gains.
- **Implication:** Sustainable surge pricing should focus on Frequent and Occasional riders, while Rare-rider retention requires non-pricing interventions.

REFERENCE AND METHODOLOGY

METHODOLOGY

1.Data & Analysis

Dataset: India Ride-Hailing Dynamic Pricing Dataset (~50,000 rides).
Analysis conducted using Python (Pandas, NumPy, Matplotlib, Seaborn) and Microsoft Excel.
Demand, cancellation, satisfaction, churn, and revenue metrics were analyzed across surge bands, zone types, cities, and time segments.

2.Elasticity Estimation

Price elasticity estimated using the Arc Elasticity (Midpoint) Method.
Ride completion rate was used as a proxy for realized demand.
Elasticity calculated separately by zone type, customer segment, and time segment.

3.Financial Simulation

Contribution Margin (CM), LTV/CAC, Breakeven Churn, and Scenario Simulations were developed in Excel.
Three pricing strategies were evaluated:
Option 1: Calibrated Pricing
Option 2: Zone-Differentiated Pricing
Option 3: Loyalty Protection

REFERENCES

Data Sources

- 1.India Ride-Hailing Dynamic Pricing Dataset (Project Dataset)
- 2.Demand, cancellation, satisfaction, and churn analyses developed in Python (Pandas, NumPy, Matplotlib, Seaborn)
- 3.Elasticity estimation report (Deliverable 2)
- 4.Unit Economics Workbook (Deliverable 3)
- 5.Revenue & Margin Scenario Simulation Workbook (Deliverable 4)

Analytical Frameworks

- Price Elasticity of Demand (PED)
- Customer Lifetime Value (LTV)
- Customer Acquisition Cost (CAC)
- Contribution Margin Analysis
- Churn Sensitivity Analysis
- Porter's Five Forces Framework

THANK YOU

